

Introduction

ZOMATO DATASET ANALYSIS

Alfido Tech Data Science Internship

Prepared By:
Prathamesh Borkar

Task Number: 1

Objective:

To analyze restaurant and review data to identify trends in ratings, cuisines, locations, and pricing, and provide actionable business recommendations.

Executive Summary

This project analyzes the Zomato restaurant dataset to identify patterns in restaurant ratings, cuisine preferences, pricing, and location popularity. The dataset was cleaned by removing duplicate records, handling missing values, converting ratings and costs into numeric formats, and filtering invalid entries.

The analysis revealed that BTM, Whitefield, HSR, and Indiranagar are among the most active restaurant locations. North Indian cuisine emerged as the most popular cuisine category, followed by Chinese and South Indian cuisines.

The relationship between restaurant price and customer ratings was found to be weak, indicating that higher-priced restaurants do not necessarily receive better customer reviews. Most restaurants fall within the affordable-to-mid-range pricing segment.

Based on the findings, recommendations have been provided to support restaurant partnerships, content creation, and customer engagement strategies for a platform similar to Alfido Tech.

Dataset & Data Cleaning

DATASET OVERVIEW

Dataset: Zomato Restaurant Dataset

Initial Dataset Size:

56,252 rows and 13 columns

Data Cleaning Performed:

- Removed 15,703 duplicate records.
- Handled missing values in important columns.
- Converted restaurant ratings into numeric values.
- Cleaned the cost column and removed unrealistic values.
- Removed invalid cuisine records containing review text.
- Prepared the dataset for analysis and visualization.

Final Dataset Size:

38,003 rows and 13 columns

TOOLS USED

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- WordCloud
- Google Colab

Visualizations & Findings

Chart 1: Top 10 Restaurant Locations

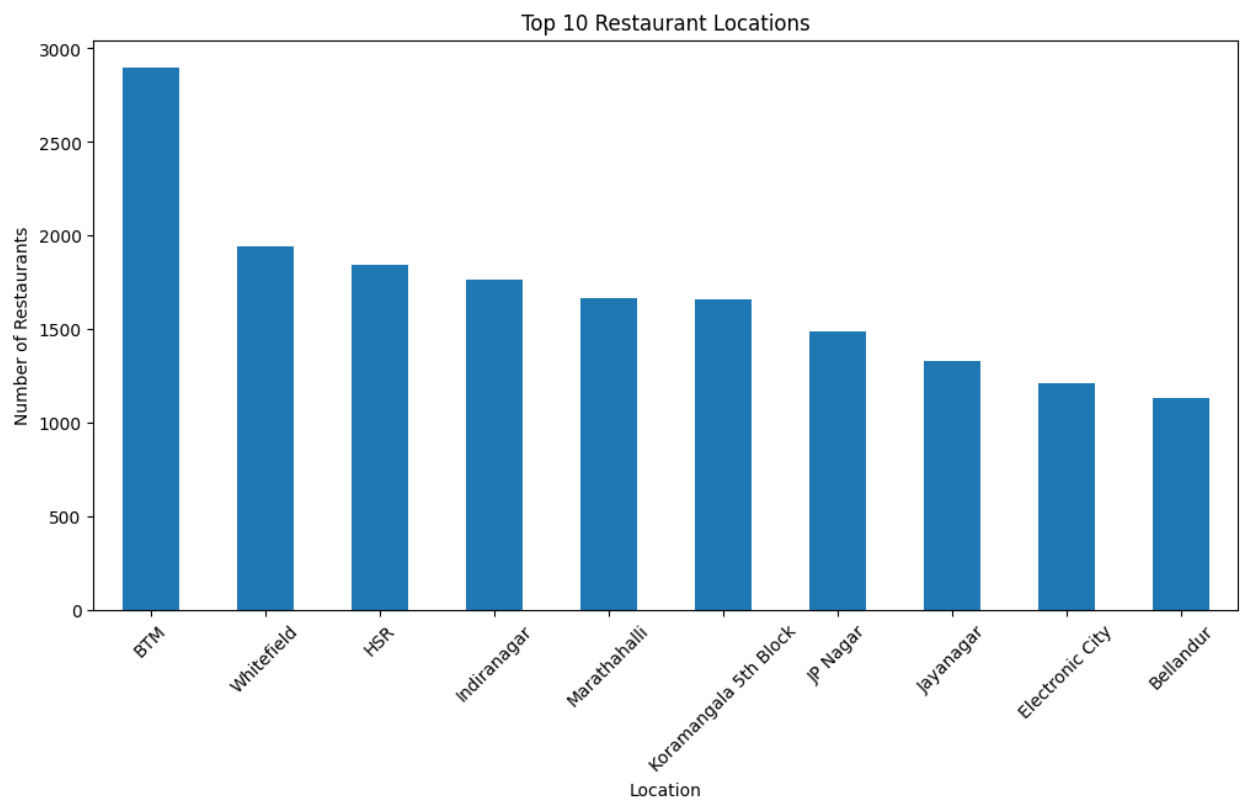


Chart 2: Top 10 Most Popular Cuisines

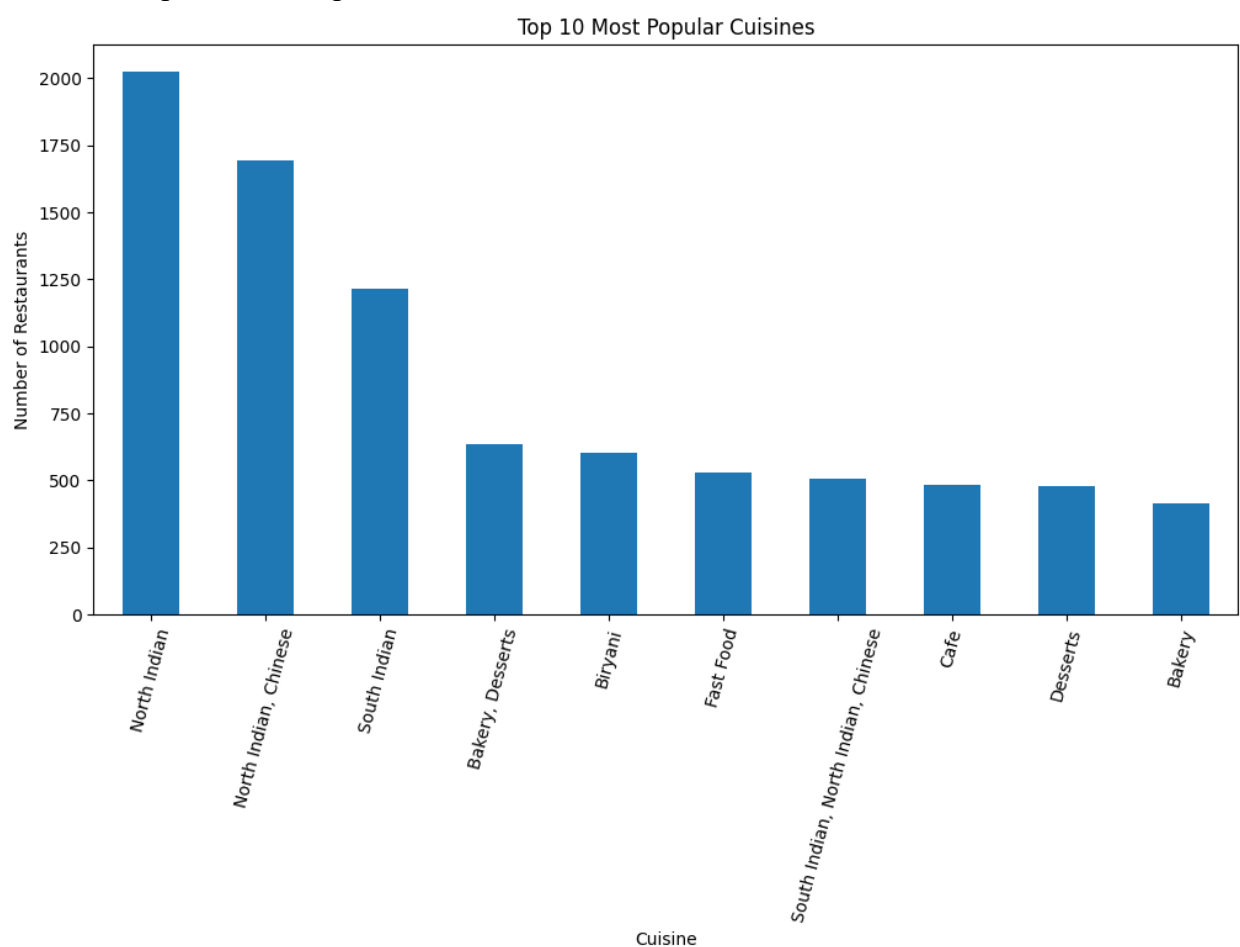


Chart 3: Price vs Rating

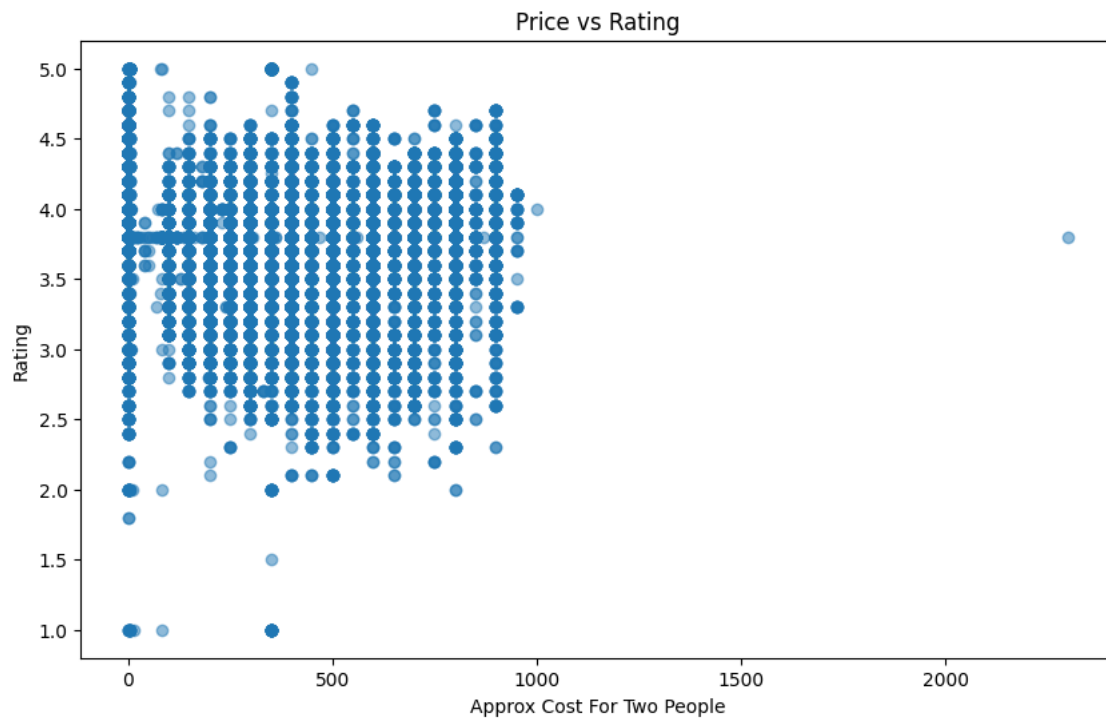


Chart 4: Correlation Heatmap

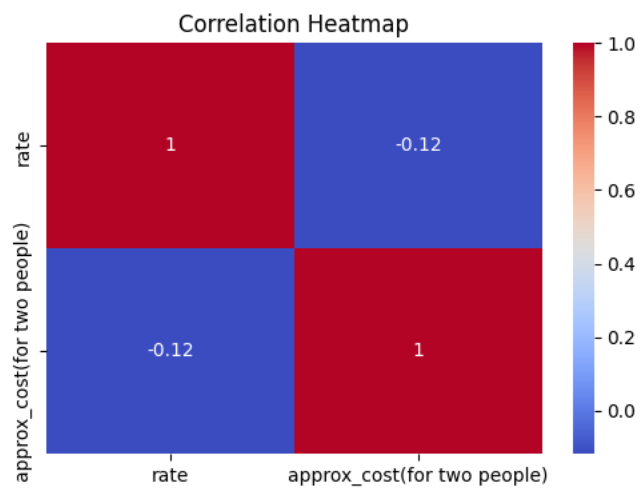


Chart 5: Cuisine Word Cloud

The word cloud displays a variety of cuisines and food items. The most prominent words are 'Indian' and 'Chinese'. Other significant words include 'Biryani', 'North', 'South', 'Fast', 'Street', 'Food', 'Desserts', 'Bakery', 'Ice Cream', 'Mughlai', 'Beverages', 'Finger', 'Pizza', 'Burger', 'Kebab', 'Rolls', 'Seafood', 'Fruit', 'Juices', 'Mithai', 'Mughi', 'Andhra', 'Kerala', 'Bengal', 'Gujarat', 'Maharashtra', 'Tamil Nadu', 'Karnataka', 'Goa', 'Odisha', 'West Bengal', 'Assam', 'Arunachal Pradesh', 'Nagaland', 'Mizoram', 'Tripura', 'Manipur'.

Key Findings

- BTM is the most restaurant-dense location in the dataset.
- Whitefield, HSR, and Indiranagar are major restaurant hubs.
- North Indian cuisine is the most popular cuisine category.
- Chinese and South Indian cuisines are also highly represented.
- Most restaurants are priced in the affordable-to-mid-range segment.
- Restaurant ratings generally fall between 3.5 and 4.5.
- Price and rating show a weak negative correlation (-0.12).

Recommendations & Conclusion

Recommendations

1. Partner with highly rated restaurants in BTM, Whitefield, and HSR.
2. Focus marketing efforts on popular cuisine categories such as North Indian and Chinese.
3. Develop personalized recommendation systems using ratings and cuisine preferences.
4. Promote mid-range restaurants, as they represent the largest market segment.
5. Create data-driven content such as "Top Restaurants by Location" and "Most Popular Cuisines"

Conclusion

The Zomato dataset analysis provided valuable insights into restaurant distribution, cuisine preferences, pricing patterns, and customer ratings. The findings indicate that restaurant success is influenced more by customer experience and food quality than by price alone. These insights can help businesses and platforms make informed decisions regarding partnerships, marketing strategies, and customer engagement initiatives.